

StudyMingle Sample Worksheet

Engineering Mechanics - First Year

Question 1

Resolve a 10 N force acting 30 degrees above the horizontal into horizontal and vertical components.

Question 2

Two perpendicular forces of 6 N and 8 N act at the same point. Find the magnitude and direction of the resultant force.

Question 3

A 20 kg box rests on a horizontal surface. Using $g = 9.81 \text{ m/s}^2$, calculate its weight and the normal reaction when no other vertical forces act.

Question 4

A car increases its velocity from 5 m/s to 25 m/s in 10 seconds. Calculate its uniform acceleration and the distance travelled during this interval.

Question 5

A simply supported beam carries a central point load of 12 kN. Determine the vertical reaction at each support.

How to use: upload this PDF to StudyMingle, review the extracted questions, choose one question, and work with the Study Coach. Try your own reasoning first, request progressive hints, or type "solve it" to receive a complete worked explanation.